

Lab 1 — Setting up Google Colab

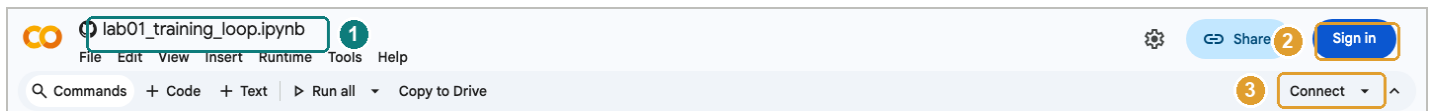
Everything runs in your browser. Nothing to install.

What you need: a web browser and a **free Google account** (a personal @gmail.com is fine — some hospital and university accounts block Colab, so if yours does, use a personal one). **You do not need:** Python, PyTorch, an install, or a GPU. *Colab* is a free Google service that runs Python notebooks on Google's computers; your laptop only shows the page.

Step 1 — Open the Lab 1 notebook

Click the Lab 1 link (on the course page, or type it in). It opens Colab with the notebook already loaded — there is nothing to download first.

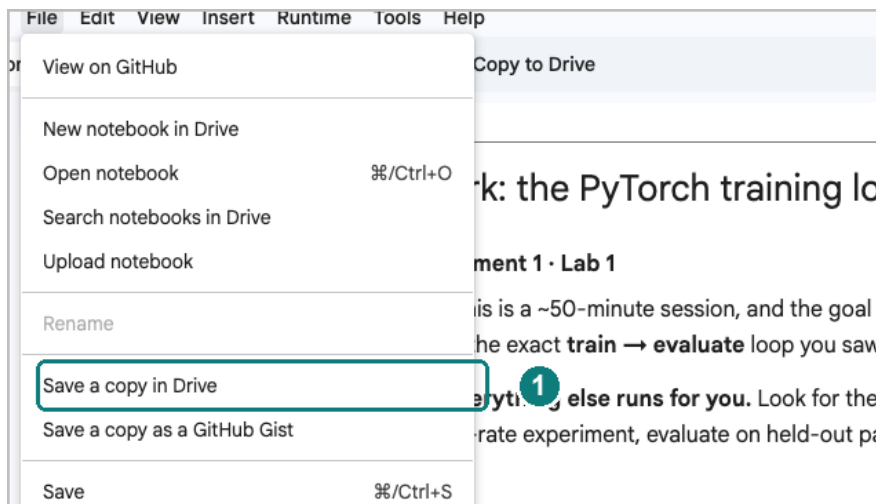
https://colab.research.google.com/github/indigobio/msacl_dl_course/blob/main/student_pack/labs/lab01_training_loop.ipynb



- (1) Check the file name reads `lab01_training_loop.ipynb` — that is the right notebook.
- (2) **Sign in** (top right) with your Google account. Until you do, you can read the notebook but not run it.
- (3) **Connect** (top right) starts your free machine. It usually connects on its own the first time you run a cell; if it says *Connect*, click it and wait for a green tick.

Step 2 — Save your own copy (do this before you type anything)

The link opens a *read-only* copy. Use **File** → **Save a copy in Drive** and work in that copy, or your edits will be lost when you close the tab. Colab opens the copy in a new tab, named `Copy of lab01_training_loop.ipynb` — **that is the tab you want to work in**.



Step 3 — Run a cell

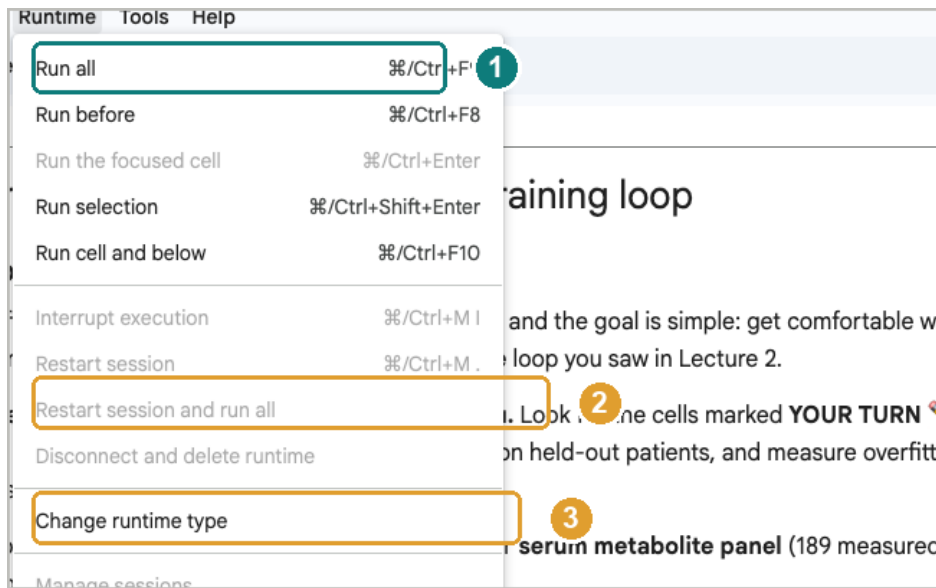
A notebook is a list of *cells*. A cell holds either text or code. Click a code cell, then press **Shift + Enter** — or hover over the (1) bracket at its left, which turns into a ► button, and click it.



- Run the cells **in order, top to bottom**. Later cells depend on earlier ones.
- While a cell runs the bracket shows a spinner; when it finishes you get a number, e.g. [1].
- The very first run installs things and fetches the data — give it a few seconds.
- Only the cells marked **YOUR TURN** (the pencil icon) need editing. Run all the others exactly as they are.

Step 4 — If something gets tangled

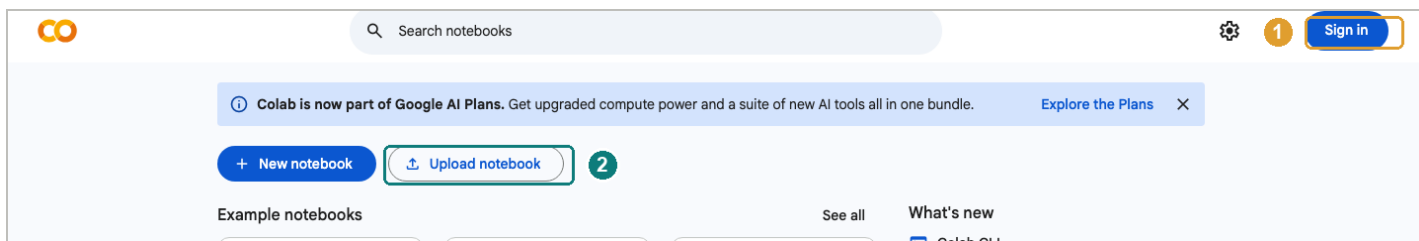
The **Runtime** menu is the fix for almost everything.



- **(1) Run all** — runs the whole notebook from the top. Good after you fix a cell.
- **(2) Restart session and run all** — throws away every variable and starts clean. This is the answer to “it worked a minute ago”. You lose nothing but the running state; your typed code stays.
- **(3) Change runtime type** — where you would pick a **T4 GPU**. **Lab 1 does not need one**; the default CPU is plenty and connects faster. You will use this in Lab 2.

Step 5 — If the link does not open

Go to colab.research.google.com directly, sign in **(1)**, then use **File** → **Open notebook** → **GitHub** and paste [indigobio/msacl_dl_course](https://github.com/indigobio/msacl_dl_course). Or use **Upload notebook** **(2)** if an instructor hands you the `.ipynb` file on a USB stick.



When things go wrong

What you see

“Sign in” keeps reappearing, or Colab will not load

`NameError / ... is not defined`

A cell has been running for minutes

“Cannot connect to runtime” / disconnected

An `assert` cell failed

You edited the wrong tab and lost your work

What to do

Your institutional Google account may block Colab. Sign in with a personal `@gmail.com` instead.

You skipped a cell, or ran them out of order. **Runtime** → **Restart session and run all**.

Normal for training cells. If it is stuck, **Runtime** → **Interrupt execution**, then restart and run all.

Free Colab disconnects after a while idle. Click **Connect** and **Runtime** → **Run all**; you lose only the running state.

That is the notebook checking your answer — it is meant to catch you. Read the message, fix the marked line, re-run that cell. The **Code Hint Sheet** lists the options for every blank.

You were in the read-only original. Redo **File** → **Save a copy in Drive** and check the tab title starts with **Copy of**.

Screenshots are of the Google Colab interface, captured 22 September 2026, with callout numbers added for this sheet. Colab is a Google product and its appearance changes from time to time — if a button has moved, the menu path in the text is still the one to follow.